

Annex A: Structural Macro Policy Simulation Model

The model in this study largely follows the relatively simple and parameter thrifty FPAS approach (Berg, Karam, & Laxton, 2006), with some attempted enrichments. The FPAS is a simplified form of a New Keynesian DSGE macroeconomic model with micro foundations and forward-looking rational expectations, though the microeconomic foundations are implicit. The model produces a baseline forecast and fiscal and monetary policy scenarios for a 12-16 quarter horizon, and also aims at facilitating coordination with NBR, which has its own quarterly projection model, also of the FPAS-type.¹

A. Model overview

The model draws on recent FPAS applications, especially the IMF-supported FPAS Central Bank models for Rwanda, Uganda, and Kenya, India (Benes, et al., 2016), South Africa (Botha, de Jager, Ruch, & Steinbach, 2017), and other recent experiences in sub-Saharan Africa (Berg & Portillo, 2018).

The model is small, open economy; has the interest rate (endogenous) as nominal anchor of the system; focuses on output and inflation under flexible exchange rates as the main policy makers' concerns; and incorporates fiscal and monetary policy which both matter in the short-run. Compared with other FPAS, it has the following new features to meet the needs for policy scenarios at the Ministry of Finance (MINECOFIN) and compare results with those of IMF-type financial programming projections:

Demand—IS curve (and goods equilibrium)²

(i) the IS equation is replaced by structural components (C, I, G, E, M) whose behavioral specifications are similar to the reduced-form of DSGE and which together determine output in the short-run.³ More explicit behavioral content in demand seems appropriate also in light of the weakness of transmission mechanism;

(ii) the output gap is formulated for total GDP even though agricultural GDP is hardly affected by the business-cycle or policy responses and is subject to its own mostly weather-related shocks, and for this reason some FPAS use a non-agricultural output gap in IS and Phillips curves.⁴ However, the total GDP output gap is a good proxy for the non-agricultural output gap which in turn affects core inflation (i.e. excluding food) in the inflation equation;⁵

¹ The model is part and parcel of MINECOFIN's toolbox, which includes a short-term leading-indicator Bridge-Equation-type now/forecasting tool; a medium-term Financial Programming-type macroeconomic framework; a medium-term multi-sector CGE and a long-term macroeconomic DGE. The latter two generate alternative paths for output growth, real interest and real exchange rates.

² The IS curve also reflects short-term savings investment equilibrium, brought about by variation in nominal income (output and prices, and underlying employment) in Keynesian fashion.

³ In parallel, the model uses a reduced-form IS curve which subsumes the five demand components.

⁴ E.g. India (Benes, et al., 2016), Rwanda (Mwenese & Kwizera, 2018).

⁵ The non-agricultural and total output gaps are strongly correlated as fluctuations in agriculture (other than normal seasonalities) are uncorrelated with the business cycle. With a weight of some 30% in GDP, agriculture dampens the business cycle of non-agricultural GDP.

(iii) government ‘fiscal stance’ affects output, while taxes affect disposable income (and demand), implying that part of households are hand-to-mouth, liquidity-constrained consumers, who do not take into account future taxes to pay for deficits. The prevalence of liquidity constrained makes monetary policy less effective and fiscal policy more relevant. The financing of the deficit is also traced: after exogenous net foreign and domestic nonbank financing, the residual is bank credit.

Supply (Phillips curve and equilibrium output)

(iv) a distinction is made between headline and core inflation (the latter excluding fresh food and energy), but also between pure domestic inflation (wages, profits) and import inflation (final demand and intermediate imports of goods and services);

(v) potential output is based on a macro production function (though not directly linked to investment decisions and intertemporal profit maximization as in DSGE). Private capital is endogenous (from the structural investment equation) and public capital derives from exogenous government investment, while the growth rates of labor, human capital and TFP are exogenous.

Exchange rate rule (and external equilibrium)

(vi) the role of the standard risk-adjusted Uncovered Interest rate Parity (UIP) is mitigated to reflect limited capital mobility, imperfect asset substitutability, and low pass-through of interest differentials in exchange rates in Rwanda.⁶ Hence an UIP-based exchange rate rule is combined with backward-looking expectations and a gradual move of the exchange rate in line with the change in the equilibrium real exchange rate and the differential between domestic and foreign inflation targets;⁷

(vii) this hybrid rule implies that the exchange rate equation has elements of exchange rate targeting—using central bank (NBR) intervention as instrument to smoothen exchange rate movements—in addition using the interest rate as instrument to target inflation, i.e., a form of dual targeting aimed at reducing exchange rate volatility;

(viii) assuming less than full UIP implies that external equilibrium is no longer assured by fully accommodating capital flows which allow the exchange rate to meet the UIP condition *and* equilibrate the balance of payments. Instead, the hybrid exchange rate rule is compatible with a net capital inflow function (of interest rate differential and expected exchange rate change)⁸ and a residual forex intervention by the NBR, for an endogenous current account deficit.

Monetary policy rule (and monetary equilibrium)

(ix) the interest-rate (Taylor) rule accommodates some exchange rate targeting, also because a weak transmission mechanism and prevalence of high spreads makes for a smaller impact of any interest rate change (deposit rates are in part administered);

(x) a money demand function and money supply are specified, but without a role in the model, but as a consistency check because Rwanda’s monetary policy is still a hybrid of targeting interest rates and

⁶ The UIP seems the least satisfactory element of FPAS to apply to LIC, given its weak empirical validation (Chin and Meredith (1998)) and the strong assumption of mobile capital. It has also difficulties in accounting for foreign trade shocks (Linde, 2018, p. 280) as the trade balance has no impact on the external balance.

⁷ A modified UIP was first applied by Benes, Vavrá and Vlcek in 2002, cited in (Benes, Hurnik, & Vavra, 2008). It has been applied in several FPAS models, e.g., Czech Republic (Al-Mashat, Clinton, Laxton, & Wang, 2018), Canada (Al-Mashat, Clinton, Laxton, & Wang, 2018), India (Benes, et al., 2016), South Africa (Botha, de Jager, Ruch, & Steinbach, 2017) and Rwanda (Charry, Gupta, & Thakoor, 2014).

⁸ See (Ostry, Ghosh, & Chamon, 2012).

money quantities. This entails tracing the flows of net credit to government and net foreign assets as components of money supply, which meet endogenous money demand.

Long-run real interest rates and exchange rates

(xi) While the equilibrium paths of real interest and exchange rates are derived from HP filters with first-order AR processes, they are in parallel obtained from a long-term DGE model that projects their equilibrium paths. The steady state inflation target is 5% (EAC convergence criterion), and the inflation target each period moves to the steady state via a first-order AR process.

Overall, the model combines standard FPAS features for IT with a role for exchange rate targeting, structural macro and fiscal aspects, and elements of financial programming of macroeconomic flows, thus addressing needs of both fiscal and monetary policy making.

B. Model specification and calibration

The model—shown in detail by four blocks corresponding to the main FPAS equations in **Box 2**—follows the standard FPAS specification with a number of new features.

The **first block** has five structural equations, one for each final demand category, thus staying closer to the underlying DSGE model. Government demand is exogenous and net taxes (including government transfers to agents) are deducted from disposable income in the consumption function, hence a fiscal expansion affects total demand and output. Atypical for FPAS, the financing of the fiscal balance is traced, but with no direct role in the system. It is broken down into foreign and domestic nonbank financing (both exogenous), and bank financing (residual). The latter is part of money supply, which the CB must adjust to match money demand at the interest rate. Output, equal to the sum of demands, is seasonally adjusted and related to potential output to generate the output gap. The five **structural equations (1.1-5)** are also combined in a reduced form (though not solved analytically) into the standard IS curve for the output gap (**equation 1.0**).

The **second block** has the inflation equations (Phillips curve, i.e. short-term supply), separated into core and non-core inflation. Atypical for FPAS, it distinguishes—in addition—between domestic and import inflation, because output and real exchange rate gaps matter mostly for purely domestic inflation. The specification seeks to avoid adding relative prices as variables—which seem to complicate certain FPAS models—and instead, define the real exchange rates as specific to the inflation rate under study, e.g. core inflation is related to core real exchange rate (with core inflation in denominator).

The **third block** determines the exchange rate “rule” as a combination of UIP and move to the equilibrium nominal exchange rate. This means that UIP does not hold, capital mobility is imperfect and financial assets are imperfect substitutes.⁹ It includes a balance of payments equation where the current account deficit (equations 3-4 and exogenous factor income and transfers) is financed by endogenous capital inflows as a function of expected return differential, and residual forex intervention by NBR. The exchange rate used is the RWF/US\$ rate, even if NBR considers a basket of

⁹ Following (Benes, et al., 2016) and (Ostry, Ghosh, & Chamon, 2012).

currencies in managing the exchange rate. As foreign prices are in US\$ at trading partners' exchange rates (realized or projected), so the model uses real *effective* exchange rates.

The **fourth block** has the Taylor interest rate “rule”. It includes money demand and *implicit* monetary equilibrium, even though monetary variables other than the interest rate play no direct role in the system. It also specifies the relationship between deposit or lending rates and the policy rate, considering term structure, with longer-term interest rates set equal to the average of expected short-term interest rates over the maturity.

The **fifth block** generates the “unobserved variables”, i.e. trend values for potential output growth, real interest rate, real exchange rate, and target inflation rate. It assumes first-order AR processes vis-à-vis the steady state levels (in quarter t) for potential output growth rate, real interest rate, log of real exchange rate and target inflation rate.

The **sixth, foreign block** is kept simple having fewer AR processes and considering foreign variables more exogenous. For projections, it assumes constant output growth rate, interest rate, commodity price inflation trade (trade, food, energy), and CPI inflation rate, which are all trade-weighted for Rwanda.

The **seventh block** states the equilibrium parity conditions. Most model variables are formulated in deviation from equilibrium levels, and in the absence of shocks all variables converge to their equilibrium path. On the equilibrium path, the relative Purchasing Power Parity (PPP) holds by definition of the real exchange rate and UIP holds up to the country risk premium. If the country risk premium is given, the domestic equilibrium real interest rate follows and vice versa.

Variable definition and notation mostly follow custom for FPAS. All variables are quarterly, and output and price variables are seasonally adjusted. Gaps are deviations from equilibrium levels, output and exchange rates are in logs, interest rates are in percent, inflation rates π are annualized in percent, other relative changes are formulated as Δ log; all variables are by quarter and changes relative to previous quarter, except for year-on-year inflation π_4 which is relative to four quarters back. Superscripts X^* denote equilibrium value, X^{ss} steady state value, X^f foreign variable, and $\bar{X}$ exogenous variable. All variables are in constant prices unless indicated otherwise, notably for nominal flows which are (converted back) to current RWF.

Very preliminary parameters for an initial model calibration—also in comparison with a few other studies or stylized facts—are shown in **Box 3**.

Box 2. Rwanda Quarterly Macro Policy Simulation model, FPAS-type

1. Demand (structural equations and IS curve)

Real consumption (**C**) function of disposable income (**YD**) and real interest rate (**RRdep**):

$$\ln(C_t) = \gamma_1 \ln(C_{t-1}) + \gamma_2 RRdep_t + \gamma_3 \ln(YD_t) + \gamma_4 RRdep_{t-1} + \gamma_5 \ln(YD_{t-1}) \quad (1.1)$$

$$RRdep_t = idep_t - E[\pi_{t+1}] \text{ real deposit rate}$$

Net in/direct taxes and transfers (**T**) and disposable income, in real terms:

$$\ln(T_t) = \tau_1 \ln(Y_t) + \tau_0 \quad (1.1a)$$

$$YD_t = Y_t - T_t \quad (1.1b)$$

Real investment (**I**) function of GDP (**Y**) and real interest rate (**RRlen**):

$$\ln(I_t) = \delta_1 \ln(I_{t-1}) + \delta_2 RRlen_t + \delta_3 \ln(Y_t) + \delta_4 RRlen_{t-1} + \delta_5 \ln(Y_{t-1}) \quad (1.2)$$

$$RRlen_t = ilen_t - \pi_{t+1}^E \text{ real lending rate}$$

Real exports (**E**) function of foreign demand (**Y^F**) and real (export) exchange rate (**z**):

$$\ln(E_t) = v_1 \ln(E_{t-1}) + v_2 z_t + v_3 \ln(Y_t^F) + v_4 z_{t-1} + v_5 \ln(Y_{t-1}^F) \quad (1.3)$$

Real imports (**M**) function of GDP and real (import) exchange rate:

$$\ln(M_t) = \mu_1 \ln(M_{t-1}) + \mu_2 z_t + \mu_3 \ln(Y_t) + \mu_4 z_{t-1} + \mu_5 \ln(Y_{t-1}) \quad (1.4)$$

Real government consumption and investment, exogenous:

$$G_t = \overline{G_t} \quad \text{and} \quad IG_t = \overline{IG_t} \quad (1.5)$$

Government balance (**FB**) net taxes minus expenditure plus net foreign transfers/interests:¹⁰

$$FB_t = T_t - (G_t + IG_t) + NFG_t = FBadj_t + FBcyc_t \quad (1.5a)$$

where: **FBadj_t** adjusted for revenue and spending due to output above/below capacity

FBcyc_t cyclical part of fiscal balance

$$NFG_t = \overline{NFG_t}^{\$} S_t / P_t^{gdp} \text{ real net foreign transfers/interest (deflated by GDP deflator)} \quad (1.5b)$$

fbcyc_t = **FBcyc_t** / **Y_t^{*}** “fiscal impulse”: cyclical part of fiscal balance over capacity output

$$= ygap_t \left(\frac{T_t}{Y_t} \xi_T - \frac{E_t}{Y_t} \xi_E \right) \quad (1.5c)$$

where: ξ_T , ξ_E elasticities of revenue and expenditure (**E_t** = **G_t** + **IG_t**) vis-a-vis output gap¹¹

Tracing deficit financing (debt):

¹⁰ Deflated by GDP deflator (**P^{gdp}**).

¹¹ The cyclical component is obtained by adjusting actual revenue and spending for the impact of the output gap (cf. Bornhorst et al (2011), requiring only elasticities of revenue and spending vis-à-vis the output gap. The revenue elasticity is expected to be close to 1, and the expenditure elasticity close to zero.

$\Delta NCG_t = FB_t P_t^{gdp} - \Delta \overline{BD}_t - \Delta \overline{BF}_t^s S_t$ deficit after domestic nonbank borrowing ($\Delta \overline{BD}_t$) and foreign borrowing is financed by bank credit (1.5d)

GDP determined by sum of final demands

$$Y_t = C_t + I_t + (G_t + IG_t) + E_t - M_t \quad (1.6)$$

Seasonally adjusted GDP and output gap

$$Y_t^{sa} = \sigma \alpha_t^s Y_t \quad (1.6a)$$

where: $\sigma \alpha_t^s$ seasonality factor for quarters $s=1...4$ if t is quarter 1...4

$$ygap_t = \ln(Y_t^{sa}/Y_t^*) \text{ output gap (seasonally adjusted) } Y_t^* \text{ potential output} \quad (1.6b)$$

Reduced-form demand (IS-curve): output gap function of real interest and exchange rate gaps, foreign demand, and fiscal impulse:

$$ygap_t = \alpha_1 ygap_{t-1} + \alpha_2 RRgap_t + \alpha_3 zgap_t + \alpha_4 ygap_t^F + \alpha_5 fbcyc_t \quad (1.0)$$

where:

$$RRgap_t = RR_t - RR_t^* = (i_t - E[\pi_{t+1}]) - (i_t^* - \pi_{t+1}^*) \text{ real interest rate gap} \quad (1.0a)$$

i and i^* nominal (equilibrium) interest rate, π^* and $E[\pi]$ target and expected inflation

$$zgap = z - z^* = \ln\left(\frac{\overline{PWS}}{P}\right) - \ln\left(\frac{\overline{PWS}}{P}\right)^* \text{ real exchange rate gap in terms of CPI (P)} \quad (1.0b)$$

In parallel, a real exchange rate is defined specific to inflation component under study, e.g.:

$$zgap^{core} = \ln\left(\frac{\overline{PWS}}{p_{core}}\right) - \ln\left(\frac{\overline{PWS}}{p_{core}}\right)^* \text{ real exchange rate gap in terms of core inflation (P}^{core}) \quad (1.0c)$$

$z^* = \ln\left(\frac{\overline{PWS}}{P}\right)^*$ equilibrium real exchange rate in terms of CPI (P) and average “world” price (of Rwanda’s trading partners) in domestic currency ($\overline{PWS}$)

$$ygap_t^F = \ln(Y_t^F/Y_t^{F*}) \text{ foreign output gap}$$

2. **Short-term supply**, expectations-augmented Phillips curve: function of output gap, real exchange rate gap (or first difference), and indirect effects of food and energy prices:

Core inflation:

$$\pi_t^{core} = \beta_1 E[\pi_{t+1}^{core}] + (1 - \beta_1) \pi_{t-1}^{core} + \beta_2 ygap_t + \beta_3 (z_t - z_{t-1}) + \beta_4 \pi_t^{food} + \beta_5 \pi_t^{energy} \quad (2.1)$$

Food and energy prices as a function of their past values and import prices:

$$\pi_t^{food} = \beta_{21} E[\pi_{t+1}^{food}] + (1 - \beta_{21}) \pi_{t-1}^{food} + \beta_{22} (\overline{pw}_t^{food} + \Delta s_t) \quad (2.2)$$

$$\pi_t^{energy} = \beta_{31} E[\pi_{t+1}^{energy}] + (1 - \beta_{31}) \pi_{t-1}^{energy} + \beta_{32} (\overline{pw}_t^{energy} + \Delta s_t) \quad (2.3)$$

where: $\pi_t = (P_t/P_{t-1} - 1) * 4$ inflation rate, vis-à-vis previous quarter, annualized

π_{t-1} inflation rate in quarter t-1, backward-looking component (inertia)

$E[\pi_{t+1}]$ expected inflation rate for quarter t+1, forward-looking component

$\Delta z_t = z_t - z_{t-1}$ real exchange rate relative change

$\Delta s_t = s_t - s_{t-1} = \ln(S_t) - \ln(S_{t-1})$ nominal exchange rate relative change

$\overline{pw}_t^{food} + \Delta s_t$ relative change in food import price in domestic currency

$\overline{pw}_t^{energy} + \Delta s_t$ relative change in energy import price in domestic currency

Headline inflation:

$$\pi_t^{cpi} = \beta_{41} \pi_t^{core} + \beta_{42} \pi_t^{food} + (1 - \beta_{41} - \beta_{42}) \pi_t^{energy} \quad (2.4)$$

In parallel, the Phillips curve is formulated for purely domestic inflation (excl. imports):

$$\pi_t^{dom} = \beta_{51} E[\pi_{t+1}^{dom}] + (1 - \beta_{51}) \pi_{t-1}^{dom} + \beta_{52} ygap_t + \beta_{53} (z_t - z_{t-1}) \quad (2.5)$$

where: $\pi_t^{dom} = \left(\frac{p_t^{dom}}{p_{t-1}^{dom}} - 1 \right) * 4$ domestic inflation, CPI adjusted for prices of direct and indirect imports, annualized

$(p_t^{dom})^{1-\mu} (PM_t)^\mu = P_t$ domestic and import component in CPI (P_t), μ import share in CPI

$PM_t = \overline{PM}_t S_t$ import price in domestic currency

$p_t^{gdp} = (p_t^{dom})^{1-\varepsilon} (PE_t)^\varepsilon$ domestic and export component in GDP deflator, ε export share in GDP

3. Exchange rate “rule”, external equilibrium: combines UIP with exchange rate target which moves to equilibrium nominal exchange rate:

Nominal exchange rate:

$$s_t = E[s_{t+1}] - \theta (i_t - i_t^F - prem_t)/4 \quad \text{UIP, quarterlized interest rate differential} \quad (3.1a)$$

$$E[s_{t+1}] = (1 - \eta) s_{t+1} + \eta [s_{t-1}^T + \frac{2}{4} (\Delta z_t^* + \pi_t^{A*} - \pi_t^{A^{F*}})] \quad \text{expected ER moves to target } s^T \quad (3.1b)$$

Combining (3.1a) and (3.1b):

$$s_t = (1 - \eta) s_{t+1} - \theta (i_t - i_t^F - prem_t)/4 + \eta [s_{t-1}^T + \frac{2}{4} (\Delta z_t^* + \pi_t^{A*} - \pi_t^{A^{F*}})] \quad (3.1)$$

where: $\pi_t^{A*} - \pi_t^{A^{F*}}$ differential target domestic inflation and equilibrium foreign inflation

i_t, i_t^F domestic and foreign policy interest rates

$s_{t-1}^T = s_{t-2}^T + \frac{1}{4} (\Delta z_{t-1}^* + \pi_{t-1}^{A*} - \pi_{t-1}^{A^{F*}})$ target exchange rate (by quarter) driven by change in ERER and inflation target differential (3.1c)

Net private capital inflows under imperfect capital mobility (in forex)¹²

¹² Following (Ostry, Ghosh, & Chamon, 2012). With partial adjustment, the capital stock converges to a finite stock, determined by the expected return differential.

$$\Delta \ln(BP_t) = \zeta_1 [(i_t - i_t^F - prem_t)/4 - (E[s_{t+1}] - s_t)] - \zeta_2 \ln(BP_{t-1}) \quad (3.2)$$

where: BP_t stock of private capital (cumulative net flow), gradual convergence to finite stock

External equilibrium:

$$\Delta NFA_t^{\$} = (E_t - M_t) \frac{P_t^{gdp}}{S_t} + \overline{NFG}_t^{\$} + \Delta \overline{BF}_t^{\$} + \Delta BP_t \quad \text{residual intervention given exchange rate rule, current account (in forex), and net capital inflows} \quad (3.3)$$

4. **Monetary policy “rule”** (Taylor rule): function of the equilibrium interest rate, output gap, inflation gap (in terms of headline CPI), and exchange rate gap:

Interest rate:

$$i_t = \lambda_1 i_{t-1} + (1 - \lambda_1) [RR_t^* + \pi_{t+4}^* + \lambda_2 (\pi_{t+4} - \pi_{t+4}^*) + \lambda_3 ygap_t + \lambda_4 (\Delta s_t - \Delta s_t^T)] \quad (4.1)$$

where: i_t nominal interest rate (key policy rate or interbank rate, annual)

$$i_t^* = RR_t^* + \pi_{t+4}^* \quad \text{equilibrium nominal interest rate (annual)} \quad (4.1a)$$

$$\pi_{t+4} - \pi_{t+4}^* \quad \text{deviation of expected/model-consistent inflation from target inflation (annual)}$$

$$\pi_{t+4}^E = \pi_{t+4} = P_{t+4}/P_t - 1 \quad \text{projected 4-quarter relative change in price level (annual)}$$

$$\pi_{t+4} = \sum_{i=1}^4 (1 + \pi_{t+i}) - 1$$

$$\Delta s_t - \Delta s_t^T \quad \text{deviation of exchange rate depreciation from target depreciation, quarterly}$$

Implied monetary equilibrium: money demand function, matched by money supply

$$\ln(MD_t/P_t) = \xi_1 \ln(MD_{t-1}/P_{t-1}) + \xi_2 \ln(Y_t) + \xi_3 idep_t + \xi_4 t \quad (4.2)$$

Money supply is implied by money demand for given interest rate, as CB undertakes to inject money, for given flows of NCG and NFA, such that money demand is met:

$$\Delta NCP_t = \Delta MD_t - \Delta NFA_t^{\$} S_t - \Delta NCG_t \quad \text{change in private sector credit (incl. OIN)} \quad (4.3)$$

$$\Delta NFA_t^{\$} S_t \quad \text{change in NFA, from external equilibrium (equation 3.3)}$$

$$\Delta NCG_t \quad \text{net credit to government, from government balance (equation 1.5d)}$$

$$\Delta NCP_t \quad \text{net private sector credit, attained through open market operations etc.}$$

4B. Lending and deposit rates (one-year), exogenous mark-up over key policy rate:

Lending rate:

$$ilen_t^4 = \psi_1 ilen_t^4 + (1 - \psi_1) (i_{t+4} + len_spread_t) \quad \text{and similar for other maturities} \quad (4.4)$$

$$i_{t+4} \quad \text{four-quarter ahead average of short-term policy rates}$$

$$len_spread_t \quad \text{lending rate spread vis-à-vis policy rate}$$

$$RRlen_t^4 = ilen_t^4 - E[\pi_{t+4}] \quad \text{real interest rate on one-year bonds/loans}$$

Real lending rate for investment weighted average over maturities of 1, 4, 8, 20 quarters:

$$RRlen_t = \varphi_1 RRlen_t^1 + \varphi_2 RRlen_t^4 + \varphi_3 RRlen_t^8 + \varphi_4 RRlen_t^{20} \quad (4.5)$$

Deposit rate:

$$idep_t = \psi_2 idep_{t-1} + (1 - \psi_2) (i_{t+4} + dep_spread_t) \quad (4.6)$$

dep_spread_t deposit rate spread vis-à-vis policy rate

Real deposit rate for consumption

$$RRdep_t = idep_t - E[\pi_{t+4}]$$

5. Long-run trends (steady state) and equilibrium levels (exogenous or HP filter)

Potential output

steady state growth potential output (annual, year τ):

$$g_\tau^* = \alpha k_\tau^* + \beta (l_\tau^* + hc_\tau^*) + (1 - \alpha - \beta) la_\tau^* + \chi kpu_\tau^* + tfp_\tau^* \quad (5.1a) \quad \text{from production function}$$

$$g_\tau^* = f[HP \text{ filter}(Y_t)] \quad (5.1b) \quad \text{from HP filter}$$

where: $l_\tau^*, hc_\tau^*, la_\tau^*, tfp_\tau^*$ annual growth rate labor, human capital, land, TFP (all exogenous):

k_τ^*, kpu_τ^* annual growth rate private and public capital stocks

annualized trend growth rate of potential output by quarter t in year τ (AR(1) process):

$$g_t^* = \rho_y g_\tau^* + (1 - \rho_y) g_{t-1}^* \quad (5.2a)$$

potential output and output gap by quarter t

$$Y_t^* = Y_{t-1}^* \exp(g_t^*/4) \quad \text{and} \quad ygap_t = \ln(Y_t^{sa}/Y_t^*) \quad (5.2b)$$

Within-year τ capital accumulation to derive annual growth rates (k_τ^*) and growth trend (k^*):

$$K_\tau^* = K_{\tau-1}^* (1 - \delta_p) + \sum_t I_t^* \quad \text{sum quarterly investments year } \tau \quad (5.3a)$$

$$\ln(K_\tau^*) = a + b t \quad b = k^* \quad \text{trend growth private capital stock} \quad (5.3b)$$

Equilibrium real interest rate move to steady state (SS) trend (from HP filter):

$$RR_t^* = \rho_R RR_t^{SS} + (1 - \rho_R) RR_{t-1}^* \quad (5.4)$$

where: RR_t^{SS} steady state real interest rate

Equilibrium real exchange rate moving to SS trend (from HP filter)

$$z_t^* = \rho_z z_t^{SS} + (1 - \rho_z) z_{t-1}^* \quad \text{equivalent to} \quad \Delta z_t^* = \rho_z \Delta z_t^{SS} + (1 - \rho_z) \Delta z_{t-1}^* \quad (5.5)$$

Where: $z_t^* = \ln(\overline{PW}_t^* S_t^*/P_t^*)$ equilibrium real exchange rate in terms of CPI (P_t^*)

Δz_t^{SS} equilibrium rate of real exchange rate depreciation (annual)

Target Inflation rate moving to **SS** trend

$$\pi_t^* = \rho_\pi \pi_\tau^{SS} + (1 - \rho_\pi) \pi_{t-1}^* \quad (5.6)$$

6. Foreign block (largely exogenous, with some AR(1) processes)

Foreign output gap

$$ygap_t^F = \ln(Y_t^F / Y_t^{F*}) = \ln(Y_{t-1}^F / Y_{t-1}^{F*}) + g_t^F - g_\tau^{F*} / 4 \quad (6.1)$$

Where: g_t^F growth rate foreign output (or demand, trading partners), quarter t

g_τ^{F*} growth rate foreign potential output (or demand, trading partners), annual year τ

Foreign interest rate

$$i_t^F = \rho_{iF} i_{t-1}^F + (1 - \rho_{iF}) i_\tau^{FSS} \quad \text{foreign policy interest rate, moving to steady state} \quad (6.2a)$$

$$RR_t^F = \rho_{RF} RR_{t-1}^F + (1 - \rho_{RF}) RR_\tau^{FSS} \quad \text{foreign real interest rate} \quad (6.2b)$$

Foreign prices and inflation (trading-partner weighted)

$\overline{pw}_t, \overline{pw}_t^{food}, \overline{pw}_t^{energy}$ relative change import prices (average, food, energy) in US\$

$$\pi 4_t^F = \rho_{\pi F} \pi 4_{t-1}^F + (1 - \rho_{\pi F}) \pi 4_\tau^{FSS} \quad \text{foreign inflation rate, moving to equilibrium} \quad (6.3)$$

7. Equilibrium parities for trend values if interest and exchange rates

Nominal exchange rate, PPP and UIP, annual

$$\Delta s_\tau^{SS} = \Delta z_\tau^{SS} + \pi_\tau^{SS} - \pi_\tau^{FSS} \quad \text{nominal depreciation=real depreciation+inflation differential PPP} \quad (7.1)$$

$$\Delta s_\tau^{SS} = i_\tau^{SS} - i_\tau^{FSS} - prem_\tau \quad \text{nominal depreciation=interest rate differential-premium UIP} \quad (7.2)$$

$$RR_\tau^{SS} = i_\tau^{SS} - \pi_\tau^{SS} \quad \text{and} \quad RR_\tau^{FSS} = i_\tau^{FSS} - \pi_\tau^{FSS} \quad \text{domestic and foreign real interest rates} \quad (7.3)$$

$$\Delta z_\tau^{SS} = RR_\tau^{SS} - RR_\tau^{FSS} - prem_\tau \quad \text{UIP in real terms} \quad (7.4)$$

If both PPP and UIP hold in steady state and risk premium is given, domestic interest rates align with foreign interest rates:

$$RR_\tau^{SS} = \Delta z_\tau^{SS} + RR_\tau^{FSS} + prem_\tau \quad (7.5)$$